iLearn Practice Exam

12BR · Chapter / Topic 17

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 17: Security

Name: _____

Class: _____

Date: _____

Time allowed: 35 minutes

TOTAL MARKS: 31

Marks obtained: ..... / 31

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 31.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 6 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).
· Only 31 marks of past-paper questions are available for this chapter, so the recommended time is 35 minutes (1 min 1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Secure Socket Layer (SSL) and Transport Layer Security (TLS) are two protocols.

(a) State **two** functions of SSL/TLS.

1 .....

.....

2 .....

.....

[2]

(b) Give **two** examples of situations where the use of SSL/TLS would be appropriate.

1 .....

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2 .....

.....

[2]

2 **(a)** Describe the purpose of quantum cryptography.

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..... [2]

(b) Outline **two** benefits and **two** drawbacks of quantum cryptography.

benefit 1 .....

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benefit 2 .....

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drawback 1 .....

.....

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drawback 2 .....

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[4]

3 **(a)** Describe the differences between symmetric key and asymmetric key cryptography.

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..... [3]

(b) Outline the process that would allow a business to use encryption keys to send a verified message to a group of people.

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..... [4]

4 Identify the **two** main protocols that form Transport Layer Security (TLS) **and** state the purpose of each.

Protocol 1 .....

Purpose .....

.....

Protocol 2 .....

Purpose .....

.....

[4]

5 **(a)** Define the following asymmetric key encryption terms:

(i) cipher text

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..... [1]

(ii) public key.

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..... [1]

(b) Outline the process that would allow an individual to use asymmetric key encryption to send a private message to an organisation.

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..... [4]

6 Secure Socket Layer (SSL) and Transport Layer Security (TLS) are two protocols.

Explain how SSL/TLS is used when client-server communication is initiated.

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..... [4]

END OF QUESTION PAPER

Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

Q	Past paper	Original question	Marks
1	Oct/Nov 2025 Paper 31	Q9 (a),(b)	4
2	May/June 2026 Paper 33	Q8 (a),(b)	6
3	May/June 2026 Paper 32	Q8 (a),(b)	7
4	Oct/Nov 2025 Paper 32	Q10	4
5	May/June 2026 Paper 31	Q8 (a),(b)	6
6	Oct/Nov 2025 Paper 33	Q7	4

Total: 31 marks